

AP Statistics Unit 1 Study Guide

Lessons 1.1–1.8: Exploring Data

Studies → categorical data → quantitative distributions → transformations

Built from:	The provided Lessons 1.1–1.8 materials, revised with all seven question screenshots shared on September 9, 2026.
Best use:	Learn the decision rules, cover the blue worked answers, then complete the seven essential practice questions.
Test goal:	Choose a correct method, calculate accurately, and interpret every result in context.

Complete test checklist

- ✓ Identify individuals, population, sample, and variables.
- ✓ Classify variables and studies correctly.
- ✓ Use frequency and relative frequency.
- ✓ Choose the correct denominator in a two-way table.
- ✓ Decide whether categorical variables are associated.
- ✓ Read bar graphs, segmented bars, and mosaic plots.
- ✓ Choose a quantitative display and describe it with SOCV.
- ✓ Calculate and choose among mean, median, range, IQR, and SD.
- ✓ Find quartiles, outliers, and read a modified boxplot.
- ✓ Compare distributions in context.
- ✓ Interpret percentiles and z -scores.
- ✓ Track shape, center, and spread through transformations.

A focused study plan

First pass	Read the lesson summaries and say each bold decision rule aloud.
Second pass	Cover the blue answers and redo every worked example.
Practice	Complete Questions 1–7. If time is short, prioritize 2, 5, 6, and 7.
Final pass	Grade with the key; redo only missed parts and explain the mistake in one sentence.

Revision Map: Your Seven Questions

The earlier guide covered the main concepts, but several steps needed explicit worked examples. This revision keeps Lessons 1.1–1.8 and the original seven-question practice set, and adds a worked review of the questions you shared. The older 1.1–1.3 guide is archived; its concepts remain in this full guide.

Your question	Coverage check and revision	Worked review
Album histograms	SOCV was covered. Added “at least” counts and a comparison where IQRs match but ranges and SDs differ.	Page 20
An eighth value	SD and resistance were covered. Added the two separate checks for unchanged range and increased SD.	Page 21
Raspberry weight	The z-score formulas were covered. Added the two-step method for matching standardized positions across distributions.	Page 24
Soccer dotplot	Percentiles and SD were covered. Added frequency-weighted mean, median positions, a contextual SD interpretation, and removing an extreme value.	Page 22
Employee hours	Histogram limits were mentioned. Added cumulative counts, quartile positions, and a possible IQR interval.	Page 23
Days absent	Quartiles were covered. Added converting quartile percentages into counts and a note about repeated values.	Page 21
Babysitting income	Unchanged z-scores were stated. Added the cancellation proof and explained why the other choices change.	Page 24

What to practice first

Cover each worked answer and explain its method before checking it. Focus first on the employee-hours cumulative counts, the soccer frequency table, and the matching-z-score calculation. Then use the original seven practice questions to check the whole unit.

Three distinctions these questions test

- A histogram can identify an interval containing a quartile without revealing its exact value.
- Equal IQRs do not require equal ranges or standard deviations.
- Adding one observation and adding a constant to every observation have different effects on SD.

Lessons 1.1–1.3: Categorical Data Foundations

Lesson 1.1: Statistical studies

The structure of a study

A statistical study collects data from a **sample** to answer a question about a **population**. A conclusion is strongest when the sample represents the population well.

Term	Meaning	Fast question
Individual	One object or person described by the data	Who or what is each row about?
Variable	A characteristic recorded for each individual	What changes from individual to individual?
Population, N	The entire group the conclusion concerns	Who do researchers want to understand?
Sample, n	The subset actually measured	Who supplied the data?
Census	Data from every individual in the population	Was everyone in that population included?

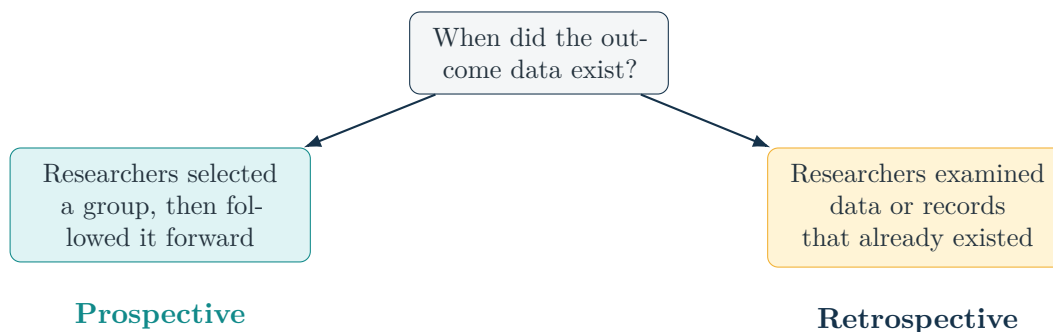
Variable type	Meaning	Examples
Categorical	Category names or group labels; arithmetic is not meaningful	eye color, car brand, yes/no
Quantitative	Numerical counts or measurements; arithmetic is meaningful	test score, time, number of siblings

For a quantitative variable, **discrete** values are countable, usually whole numbers; **continuous** values are measured and can take any value in an interval. A city population is discrete; body temperature is continuous even if rounded.

Numbers can be labels

ZIP codes, jersey numbers, and student IDs are categorical because averaging them has no useful meaning. Name a variable as what is recorded for *each individual*: “reaction time in milliseconds,” not just “reaction.”

Prospective or retrospective?



Valid limitations include unrepresentative sampling, inaccurate self-reports, a short observation period, and confounding. Observational evidence supports *association*, not cause and effect.

Safe conclusion language

Students who used one method **tended to have higher scores**, or method and score **were associated**. Do not say the method *caused* higher scores unless a well-designed experiment justifies it.

Lesson 1.2: One and two categorical variables

Display	Use	Rule
Bar graph	Frequencies or relative frequencies for categories	Bars are separated; a frequency axis begins at 0.
Pie chart	Each category's share of one whole	Categories must be mutually exclusive and exhaustive, so the slices add to exactly 100%.

Frequency language

$$\text{relative frequency} = \frac{\text{category count}}{\text{total count}}, \quad \text{percent} = 100\%(\text{relative frequency}).$$

Run it backward with

$$\text{count} = (\text{proportion})(\text{total}), \quad \text{total} = \frac{\text{count}}{\text{proportion}}.$$

From the notes, if 9 students are 30% of a sample, then $n = 9/0.30 = 30$.

Pie-chart check

You may make a pie chart only when every individual belongs to exactly one category and the categories cover the entire group. The true percentages sum to 100%; displayed rounded percentages may differ slightly. Overlapping or incomplete categories do not form a valid pie chart.

Two-way tables: choose the denominator first

	B_1	B_2	Total
A_1	a	b	$a + b$
A_2	c	d	$c + d$
Total	$a + c$	$b + d$	n

Type	Clue	Denominator
Marginal	One variable only	Grand total n
Joint	Two conditions joined by “and”	Grand total n
Conditional	“of those who,” “among,” or “given”	Total of the named given group

Same cell, different question

$$P(A_1) = \frac{a+b}{n}, \quad P(A_1 \text{ and } B_1) = \frac{a}{n}, \quad P(A_1 | B_1) = \frac{a}{a+c}.$$

Underline the group after “given” or “of those who” before dividing.

Favorite elective	Math	English	Total
Art	2	4	6
Music	5	4	9
P.E.	4	3	7
World Language	1	3	4
Technology	4	0	4
Total	16	14	30

Percent of all students who chose P.E.:

$$7/30 = 23.3\%$$

Percent who chose Math and Art:

$$2/30 = 6.7\%$$

Percent who chose Technology, given Math:

$$4/16 = 25\%$$

Percent who chose Math, given Technology:

$$4/4 = 100\%$$

Reverse condition

$P(\text{Technology} | \text{Math})$ and $P(\text{Math} | \text{Technology})$ generally use different denominators. Also, a numerator may span several cells: “private colleges” may require adding private nonprofit and private for-profit.

Lesson 1.3: Association and categorical displays

The **explanatory variable** may help explain or predict the **response variable**. To test for an association, compare the response distribution *within each explanatory group*.

How to decide whether two categorical variables are associated

1. Choose the explanatory groups.
2. Calculate the response percentages within each group.
3. Compare corresponding conditional percentages.
4. If the distributions differ meaningfully, state the association and give a specific contextual comparison.

Strong association sentence

There **is an association** between core-class preference and elective preference because the conditional distribution of elective preference differs by core-class group. For example, **25% of Math-preferring students chose Technology**, compared with **0% of English-preferring students**.

Display	What it preserves	What to compare
Side-by-side bars	Separate response bars within each group	Corresponding relative-frequency heights
Segmented bars	Equal-width, 100%-tall group bars	Segment heights; counts across groups stay hidden
Mosaic plot	Heights are conditional proportions; widths show group sizes	Heights for association, widths for group size, areas for overall counts

Mosaic plot meanings

height = conditional proportion,
width = group's share of the sample,
area = joint share of all individuals.

If a group contains five times as many individuals, its mosaic bar is five times as wide.

Teachers and students voted for a mascot. There were 100 teachers and 500 students.

	Rams	Falcons	Prairie Dogs	Group size
Teachers	80%	10%	10%	100
Students	30%	60%	10%	500
Counts	$80 + 150 = 230$	$10 + 300 = 310$	$10 + 50 = 60$	600

Although Rams have the larger teacher percentage, Falcons win overall because the student group is much larger. Use count = (conditional proportion)(group size).

Percent is not count

A larger percentage need not mean a larger count when group sizes differ. Segmented bars hide group sizes; a mosaic plot reveals them through widths. Without group sizes, do not compare counts across different segmented bars.

Misleading displays may truncate a bar graph's vertical axis, scale pictures in both height and width, use unequal intervals, or invite invalid count comparisons. A corrected frequency bar graph begins at 0.

Lesson 1.4: Describing Quantitative Data

Choose a display that matches the question

Dotplots and stem-and-leaf plots preserve individual values. Histograms group values into intervals and make the overall shape easier to see. A stem-and-leaf plot must include a key, such as $5 \mid 2 = 52$.

Display	Best feature	Limitation
Dotplot	Exact values, clusters, gaps, and possible outliers	Crowded for large data sets
Stem-and-leaf	Exact ordered values and overall shape	Awkward with many values or wide scales
Histogram	Shape of a large quantitative distribution	Exact observations are hidden; bars touch

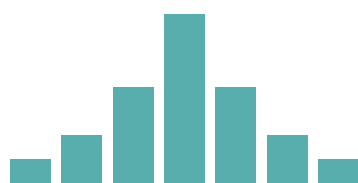
Bar graph or histogram?

A bar graph is for categorical data and its bars are separated. A histogram is for quantitative data grouped into intervals and its bars touch. The horizontal order in a histogram is numerical and cannot be rearranged.

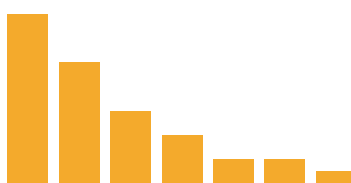
SOCV: the complete description

Always describe a quantitative distribution with **Shape, Outliers, Center, Variability**, in context.

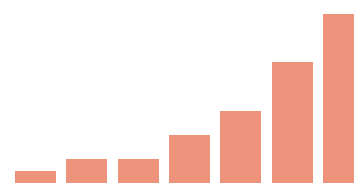
Letter	What to report	Vocabulary / measure
S	Overall shape	symmetric; skewed left/right; uni-modal, bimodal, or uniform; clusters and gaps
O	Unusual values	possible outliers, with approximate values
C	Typical value	mean when symmetric/no strong outliers; median when skewed/outliers
V	Amount of spread	range, IQR, or SD, with units



roughly symmetric



skewed right



skewed left

Tail direction names the skew. A few large values create a right tail; a few small values create a left tail.

Model SOCV description

The backpack-weight distribution is **roughly symmetric and unimodal**, with **no obvious outliers**. Its center is about **6.5 pounds** and its values range from **4 to 10 pounds**, a range of **6 pounds**.

Do not give a number without context

Write “The median backpack weight is 6.5 pounds,” not merely “the center is 6.5.” For a graph, use approximate language when exact values are not available.

Lesson 1.5: Measuring Center and Variability

Core measures

$$\bar{x} = \frac{\sum x_i}{n},$$

$$\text{range} = \max - \min,$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}},$$

$$\text{variance} = s^2.$$

The sample standard deviation s measures the typical distance of the observations from the sample mean $\bar{x}$.

Correct SD interpretation

For the five reported weekly work times, $s = 9.86$ hours means the parents' reported work times **typically** vary by about 9.86 hours from the mean of 43.6 hours.

What SD is not

SD is not the average value, not the range, and not a statement that every observation lies exactly s units from the mean. It is never negative and is 0 only when all values are equal.

Choose matching measures

Distribution	Center	Spread
Roughly symmetric with no strong outliers	Mean $\bar{x}$	Standard deviation s
Skewed or containing outliers	Median	IQR

Resistance

The **median** and **IQR** are resistant: a few extreme values have limited effect. The **mean**, **SD**, and **range** are nonresistant: an extreme value can change them substantially.

Shape	Skewed left	Symmetric	Skewed right
Mean vs. median	mean < median	mean $\approx$ median	mean > median
Reason	low tail pulls mean left	balanced tails	high tail pulls mean right

For the provided rainfall data, mean = 34.94, median = 40.5, and $s = 15.52$. The mean below the median supports the left skew visible in the class graph: small observations pull the mean toward the left tail. Mean versus median is a useful clue, not a proof of shape by itself.

Reasoning about SD without calculation

SD is larger when observations lie farther from the mean. The sets

$$A = \{40, 45, 50, 55, 60\}, \quad B = \{46, 48, 50, 52, 54\}$$

have the same mean and median, but A has the larger SD because its values are more spread out.

If one observation equal to the current mean is added, the mean stays the same and the sample SD decreases when the original SD is positive: the sum of squared deviations stays the same while the denominator increases. If all values were already equal, SD remains 0. See page 21 for adding an observation that keeps the range fixed but increases SD.

Lesson 1.6: Comparing Quantitative Data

Quartiles and the five-number summary

The five-number summary is

$$\min, \quad Q_1, \quad \text{median}, \quad Q_3, \quad \max.$$

Q_1 is the median of the lower half and Q_3 is the median of the upper half. Sort first. Under the class convention, exclude the overall median when n is odd; split evenly when n is even. Average the two middle values of any half with an even number of observations. Each quartile section contains about 25% of the observations; repeated values can make strict “below” or “above” counts differ. See pages 23 and 21.

IQR and the $1.5 \times IQR$ outlier rule

$$IQR = Q_3 - Q_1$$

$$\text{lower fence} = Q_1 - 1.5(IQR), \quad \text{upper fence} = Q_3 + 1.5(IQR).$$

Any observation below the lower fence or above the upper fence is an outlier by this rule. A second rule from the notes flags values below $\bar{x} - 2s$ or above $\bar{x} + 2s$.

For the lacrosse goals data, $Q_1 = 6$, median = 13, $Q_3 = 32$, mean = 27, and $s = 35.12$.

$$IQR = 26, \quad 6 - 1.5(26) = -33, \quad 32 + 1.5(26) = 71.$$

Thus 107 and 121 are high outliers. The $2s$ interval is

$$27 \pm 2(35.12) = [-43.24, 97.24],$$

which flags the same two values.

Modified boxplots

A modified boxplot draws the box from Q_1 to Q_3 , a line at the median, whiskers to the most extreme *non-outliers*, and separate points for outliers.



Whiskers do not always reach min and max

In a modified boxplot, the whiskers stop at the smallest and largest non-outliers. Plot each outlier separately. Boxplots show quartiles and outliers well, but they hide individual values, clusters, and gaps that a dotplot can reveal.

Compare distributions with SOCV

Use comparative words such as *higher*, *lower*, *more variable*, and *less variable*. Do not describe each group in isolation.

Model comparison

Both calorie distributions are roughly symmetric with no apparent outliers. Sandwiches have a higher median, about 830 calories, versus about 475 for salads. Sandwich calories are also more variable, with an IQR of about 340 calories compared with about 180 calories for salads.

Histogram precision

A histogram usually does not reveal the exact median or quartiles because values are grouped into intervals. It may support an interval estimate, such as “ Q_1 lies between 40% and 50%.”

Lesson 1.7: Location in a Distribution**Percentiles**

The p th percentile is a value with about $p\%$ of observations at or below it.

Percentile of an observed value

$$\text{percentile} = \frac{\# \text{ observations at or below the value}}{n}(100\%).$$

Use the counting convention from class. A percentile is a relative position, not the percent correct on a test.

If 7 of 16 recorded sleep times are at or below 119 minutes, then 119 minutes is at the

$$\frac{7}{16}(100\%) = 43.75\% \approx 44\text{th percentile.}$$

Correct percentile interpretation

A child at the 65th percentile for weight weighs at least as much as about 65% of children of the same age and sex. It does not mean the child weighs 65% more or that the weight is 65% of a maximum.

Standard scores (z-scores)**Standardized location**

$$z = \frac{x - \bar{x}}{s}, \quad x = \bar{x} + zs.$$

$z > 0$ means above the mean, $z < 0$ means below the mean, and $|z|$ is the distance from the mean in standard deviations.

For a quiz with mean 80 and SD 10, a score of 65 has

$$z = \frac{65 - 80}{10} = -1.5.$$

Interpretation: the score of 65 is 1.5 standard deviations below the mean.

Comparing different distributions

Compare standardized standing with z -scores: a higher z is farther above its own mean in SD units. The value farther from its mean has the larger $|z|$. Across differently shaped distributions, equal z -scores need not mean equal percentiles. See page 24 for finding a raw value with a matching z -score.

From the salary homework, a business salary of \$72,000 has

$$z = \frac{72 - 69.929}{5.954} \approx 0.35,$$

while a communications salary of \$65,000 has

$$z = \frac{65 - 56.467}{6.632} \approx 1.29.$$

The \$65,000 communications salary is lower in raw dollars but higher relative to its major.

Sign vs. distance

To find which value is *higher relative to its group*, compare signed z -scores. To find which is *farther from the mean*, compare absolute values $|z|$.

Lesson 1.8: Linear Transformations

Suppose every observation is transformed by

$$y = a + bx.$$

Transformation rule

Adding or subtracting changes location but not spread. Multiplying or dividing changes both location and spread. For the positive multipliers used in these lessons, the distribution's shape stays the same.

Feature	Add a	Multiply by $b > 0$
Shape	unchanged	unchanged
Mean, median, quartiles	add a	multiply by b
Range, IQR, SD	unchanged	multiply by b
Variance	unchanged	multiply by b^2
z -scores	unchanged	unchanged
Outlier status	unchanged	unchanged

In full generality, spreads are multiplied by $|b|$; if $b < 0$, the order reverses and the shape is reflected.

Summary statistics under $y = a + bx$

$$\bar{y} = a + b\bar{x}, \quad \text{median}(y) = a + b \text{median}(x), \quad s_y = |b|s_x,$$

$$IQR_y = |b|IQR_x, \quad \text{Var}(y) = b^2 \text{Var}(x).$$

Apply the addition to center measures only; do not add it to SD, range, or IQR.

Worked taxi transformation

Taxi cost is $C = 5 + 3.25d$, where d is distance in miles. The distance summary has $\bar{d} = 6.75$, $s_d = 6.61$, $Q_1 = 2.5$, median = 3.5, and $Q_3 = 10.5$.

$$\begin{aligned}\bar{C} &= 5 + 3.25(6.75) = \$26.94, \\ s_C &= 3.25(6.61) = \$21.48, \\ Q_{1,C} &= 5 + 3.25(2.5) = \$13.125, \\ Q_{3,C} &= 5 + 3.25(10.5) = \$39.125.\end{aligned}$$

The original $IQR = 8$, so the upper distance fence is $10.5 + 1.5(8) = 22.5$ miles. A 23-mile ride is an outlier. Its cost is $5 + 3.25(23) = \$79.75$, and the transformed upper fence is $5 + 3.25(22.5) = \$78.125$, so it remains an outlier.

Standardizing a distribution

Standardizing transforms every value into its z -score:

$$z = \frac{x - \bar{x}}{s} = -\frac{\bar{x}}{s} + \frac{1}{s}x.$$

The standardized distribution has the **same shape**, **mean 0**, and **SD 1**. A positive unit conversion also leaves every z -score unchanged.

Variance squares the multiplier

If SD is multiplied by 12, variance is multiplied by $12^2 = 144$. For egg sales with SD 3 dozen, variance is 9 dozen². Converting to individual eggs gives SD 36 eggs and variance 1296 eggs².

Master Decision and Formula Sheet

When the question asks...	Do this
Category share of a whole	Relative frequency; use a pie chart only if mutually exclusive, exhaustive shares total 100%.
“of all” with one category	Marginal proportion; divide by grand total.
“of all” with “and”	Joint proportion; divide by grand total.
“among,” “given,” “of those who”	Conditional proportion; denominator is the named group.
Whether categorical variables are associated	Compare complete conditional distributions; cite one specific difference.
Whether counts differ across percentage bars	Need group sizes; multiply proportion by group size.
Describe one quantitative distribution	SOCV, in context.
Choose center and spread	Symmetric/no outlier: mean and SD. Skewed/outlier: median and IQR.
Find outliers	Calculate IQR, fences, then compare every extreme value to the fences.
Draw a modified boxplot	Box at Q_1, Q_3 , median line, whiskers to non-outliers, outliers as points.
Compare quantitative groups	Compare shape, unusual features, center, and variability using comparative words and units.
Find relative position	Percentile for percent at/below; z-score for SDs above/below mean.
Transform every value	Apply $a + bx$ to center; multiply spread by $ b $; multiply variance by b^2 .

Categorical data

$$\text{relative frequency} = \frac{\text{count}}{\text{total}}$$

$$\text{count} = (\text{proportion})(\text{group size})$$

$$P(A | B) = \frac{\#(A \text{ and } B)}{\#B}$$

Quantitative data

$$\bar{x} = \frac{\sum x_i}{n}, \quad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

$$IQR = Q_3 - Q_1$$

$$Q_1 - 1.5IQR, \quad Q_3 + 1.5IQR$$

$$z = \frac{x - \bar{x}}{s}, \quad x = \bar{x} + zs$$

Seven Essential Practice Questions

These are the highest-yield question types across the provided materials. Complete them without looking at the answer key.

1. Study design and variables

Researchers randomly select 400 juniors from a school district. At the beginning of the semester, students report their main study method (flashcards, practice tests, or videos). At the end of the semester, researchers record each student's final exam score.

- Identify the population and sample.
- Identify both variables and classify each as categorical or quantitative.
- Is the study prospective or retrospective? Explain.
- State one limitation and write a conclusion that does not claim causation.

2. Two-way table, association, and displays

The results for another sample are shown below.

Study method	Passed	Did not pass	Total
Flashcards	36	12	48
Practice tests	54	6	60
Videos	20	12	32
Total	110	30	140

- Find the marginal percent who used practice tests.
 - Find the joint percent who used videos and did not pass.
 - Find the percent who passed, given flashcards.
 - Find the percent who used flashcards, given that they passed.
 - Is there an association between study method and passing? Support your answer with conditional percentages.
 - Could study method be displayed in a pie chart? State the exact condition that makes your answer valid.
 - In a mosaic plot, give the width ratio for the three study-method bars.
-

3. Display and SOCV

The number of late assignments for 10 students is

4, 4, 5, 6, 6, 6, 7, 8, 9, 14.

- Is this variable categorical, quantitative discrete, or quantitative continuous?
 - Find the mean, median, and range.
 - Find Q_1 , Q_3 , and IQR, then use the $1.5 \times IQR$ rule to check 14.
 - Describe the distribution with SOCV. Which center and spread would you report?
 - Which display, a dotplot or histogram, better preserves every individual value?
-

4. Standard deviation and resistance

Five parents report weekly work times of

38, 45, 60, 40, 35 hours.

The mean is 43.6 hours and the sample SD is 9.86 hours.

- Interpret the standard deviation in context.
 - If 60 is replaced by 120, predict what happens to the mean, median, range, and SD. Explain briefly.
 - After that replacement, which pair should summarize the data: mean and SD, or median and IQR?
-

5. Outliers and a modified boxplot

For lacrosse goals, $Q_1 = 6$, median = 13, $Q_3 = 32$, mean = 27, and $s = 35.12$. The largest observations are 50, 51, 107, and 121; the minimum is 1.

- Use the $1.5 \times IQR$ rule to identify high outliers.
 - Use the $2s$ rule to identify high outliers.
 - State the endpoints of the whiskers and list the separately plotted points in a modified boxplot.
 - Predict the shape and explain why the mean is much larger than the median.
-

6. Percentiles and z-scores

- a. In a list of 20 ordered scores, 15 are at or below Maya's score. Find and interpret Maya's percentile.
 - b. Quiz A has mean 80 and SD 10. Find and interpret the z -score for 65.
 - c. Quiz B has mean 72 and SD 6. Find the z -score for 81. Which student has the higher relative standing: the 65 on Quiz A or the 81 on Quiz B?
 - d. On Quiz A, what raw score has $z = -0.3$?
 - e. Two values in the same distribution have z -scores 1.4 and -2.1 . Which is at the higher percentile? Which is farther from its mean?
-

7. Linear transformation

Taxi cost is $C = 5 + 3.25d$, where distance d is in miles. For a set of rides,

$$\bar{d} = 6.75, \quad s_d = 6.61, \quad Q_1 = 2.5, \quad \text{median} = 3.5, \quad Q_3 = 10.5,$$

and the longest ride is 23 miles.

- a. Find the mean and SD of the costs.
 - b. Find Q_1 , the median, Q_3 , and IQR of the costs.
 - c. Use the distance summary to decide whether 23 miles is an outlier. Then verify that its cost, \$79.75, is also an outlier on the cost scale.
 - d. What happens to the shape and all z -scores?
 - e. If another variable with variance 9 is multiplied by 12, what is its new variance?
-

Answer Key and Explanations

1. Study design and variables

Answer

- Population: all juniors in the school district. Sample: the 400 randomly selected juniors.
- Main study method is categorical; final exam score is quantitative.
- Prospective, because students were selected and reported their method before researchers recorded a future exam outcome.
- One limitation is that study method is self-reported or that other variables, such as previous achievement, may confound the relationship. Safe conclusion: in this sample, study method was associated with final exam score; students using one method tended to score differently. The observational study does not prove the method caused the difference.

2. Two-way table, association, and displays

Answer

- $60/140 = 0.4286 = 42.9\%$.
- $12/140 = 0.0857 = 8.6\%$.
- $36/48 = 0.75 = 75\%$.
- $36/110 = 0.3273 = 32.7\%$. Same cell, different denominator from part (c).
- Yes. Pass rates are flashcards $36/48 = 75\%$, practice tests $54/60 = 90\%$, and videos $20/32 = 62.5\%$. The conditional distribution of passing differs by method. This is association, not proof of causation.
- Yes, if each student selected exactly one of the three methods and the three categories cover the full sample. Then their shares are mutually exclusive and exhaustive and add to 100%.
- $48 : 60 : 32$, which simplifies to $12 : 15 : 8$. Mosaic widths are proportional to group sizes.

3. Display and SOCV

Answer

- Quantitative discrete, because this is a count.
- $\bar{x} = 69/10 = 6.9$ assignments; median = $(6 + 6)/2 = 6$; range = $14 - 4 = 10$.
- $Q_1 = 5$, $Q_3 = 8$, so $IQR = 3$. The fences are $5 - 1.5(3) = 0.5$ and $8 + 1.5(3) = 12.5$. Since $14 > 12.5$, 14 is a high outlier.
- The distribution is unimodal and skewed right, with a high outlier at 14. Its center is about 6 late assignments. Because of the skew and outlier, report median = 6 and $IQR = 3$ assignments.
- A dotplot preserves every individual value.

4. Standard deviation and resistance

Answer

- The reported weekly work times typically vary by about 9.86 hours from their mean of 43.6 hours.
- The mean increases because the total increases; the sorted data become 35, 38, 40, 45, 120, so the median remains 40. The range increases from $60 - 35 = 25$ to $120 - 35 = 85$, and SD increases because 120 lies much farther from the mean.
- Median and IQR, because the extreme value makes the distribution right-skewed and those measures are resistant.

5. Outliers and a modified boxplot

Answer

- $IQR = 32 - 6 = 26$. The upper fence is $32 + 1.5(26) = 71$, so 107 and 121 are high outliers. The lower fence is -33 .
- $\bar{x} + 2s = 27 + 2(35.12) = 97.24$, so 107 and 121 are high outliers. The lower cutoff is -43.24 .
- Left whisker at 1 and right whisker at 51; plot 107 and 121 separately.
- Strongly skewed right. The two very large goal totals pull the nonresistant mean upward, while the resistant median changes much less.

6. Percentiles and z-scores

Answer

- $15/20 = 75\%$, so Maya is at the 75th percentile: about 75% of the scores are at or below hers.
- $z = (65 - 80)/10 = -1.5$. The score is 1.5 SD below the Quiz A mean.
- $z = (81 - 72)/6 = 1.5$. The Quiz B student has the higher relative standing because $1.5 > -1.5$.
- $x = 80 + (-0.3)(10) = 77$.
- The value with $z = 1.4$ is at the higher percentile. The value with $z = -2.1$ is farther from its mean because $|-2.1| > |1.4|$.

7. Linear transformation

Answer

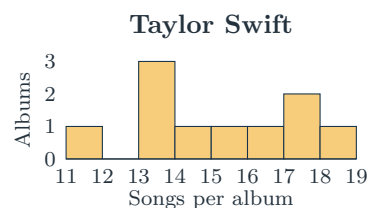
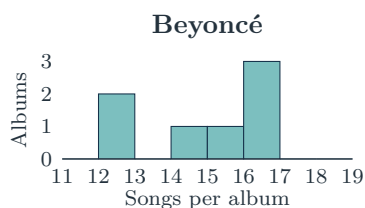
- a. $\bar{C} = 5 + 3.25(6.75) = \$26.9375 \approx \$26.94$; $s_C = 3.25(6.61) = \$21.4825 \approx \21.48 . Do not add \$5 to SD.
- b. $Q_1 = 5 + 3.25(2.5) = \$13.125$; median $= 5 + 3.25(3.5) = \$16.375$; $Q_3 = 5 + 3.25(10.5) = \$39.125$; $IQR = 3.25(8) = \$26$.
- c. Distance upper fence: $10.5 + 1.5(8) = 22.5$ miles, so 23 miles is an outlier. Cost upper fence: $39.125 + 1.5(26) = \$78.125$, and $\$79.75 > \78.125 , so the transformed value is also an outlier.
- d. The positive linear transformation preserves the shape and every z -score.
- e. $12^2(9) = 144(9) = 1296$. Variance uses the square of the multiplier.

Worked Review: Your Recent Questions

Album histograms: counts and SOCV

For the albums in your screenshot, the frequencies are:

Songs per album	11	12	13	14	15	16	17	18	Total
Beyoncé	0	2	0	1	1	3	0	0	7
Taylor Swift	1	0	3	1	1	1	2	1	10



Reading a threshold

At least 15 means $x \geq 15$, so include 15 itself. Add the *frequencies*, not the song counts: Beyoncé has $1 + 3 = 4$ albums; Taylor Swift has $1 + 1 + 2 + 1 = 5$. Together, $4 + 5 = 9$ albums have at least 15 songs.

Artist	Mean	Median	SD	IQR	Range
Beyoncé	14.429	15	1.813	$16 - 12 = 4$	$16 - 12 = 4$
Taylor Swift	14.7	14.5	2.263	$17 - 13 = 4$	$18 - 11 = 7$

A complete comparison in context

Shape: Beyoncé's song counts are concentrated at the high end with a left tail and no albums at 13 songs. Taylor Swift's distribution is roughly balanced overall, with local peaks at 13 and 17 songs.

Outliers: Neither has outliers by the $1.5 \times IQR$ rule. Beyoncé's fences are $12 - 6 = 6$ and $16 + 6 = 22$; Taylor Swift's are $13 - 6 = 7$ and $17 + 6 = 23$. All observations lie within their respective fences.

Center: The medians are close: Beyoncé's is slightly higher, at 15 songs versus 14.5. Taylor Swift's mean is slightly higher, so name the measure being compared.

Variability: Both IQRs are 4 songs. Taylor Swift has the larger range (7 versus 4 songs) and SD (2.263 versus 1.813 songs).

Remember: Equal IQRs describe equal spread of the middle 50%; the rest of the data can still produce different ranges and SDs.

Adding one value: unchanged range, increased SD

The original values are 13, 23, 10, 3, 16, 5, 7, with $\bar{x} = 11$ and $s \approx 6.95$. The choices for an eighth value are 9, 11, 21, and 25.

Two separate checks

1. Keep the range fixed: the new value must lie between the current minimum 3 and maximum 23, inclusive. Eliminate 25.
2. Check SD: adding a value close to the mean tends to reduce spread; a sufficiently distant value can increase it. Confirm using the *new* mean and sample SD.

Added value	New mean	New range	New sample SD	Meets both conditions?
9	10.75	20	6.48	No: SD decreases
11	11.00	20	6.44	No: SD decreases
21	12.25	20	7.34	Yes
25	12.75	22	8.12	No: range increases

Answer

C) 21. The range stays $23 - 3 = 20$. The new mean is $(77 + 21)/8 = 12.25$, and

$$s_{\text{new}} = \sqrt{\frac{377.5}{8-1}} \approx 7.34 > 6.95.$$

Here 377.5 is the sum of squared deviations of the eight values from 12.25.

One added observation is not a shift of every value

Appending one value changes n and can change SD. Adding the same constant to *every* value leaves SD unchanged. Also, *above the mean* alone does not guarantee an increase in SD.

Quartiles into counts: days absent

For 300 students, $Q_1 = 2$, median = 3.5, and $Q_3 = 6$ days. Since $75/300 = 25\%$, we want the statement describing one quarter of the students.

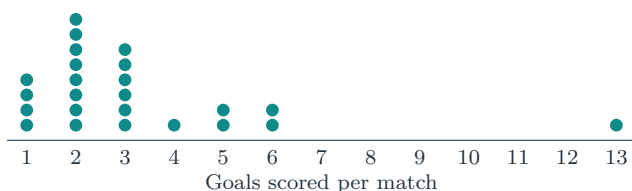
The intended answer is C) **more than 6 days**: the upper quarter lies above the third quartile, giving approximately $0.25(300) = 75$ students.

Region	Approximate share	Count out of 300
Below the median	50%	150
Between Q_1 and Q_3	50%	150
Above Q_3	25%	75

Precision note: These are the question's approximate quartile interpretations. If many students have exactly 6 absences, $Q_3 = 6$ alone does not determine the exact number *strictly above* 6. The mean does not divide the observations into fixed percentages.

Soccer dotplot: percentile, mean, median, and SD

Count every dot, including repeats. Check that the frequencies total the stated 24 matches.



Goals, x	1	2	3	4	5	6	13
Frequency, f	4	8	6	1	2	2	1
Cumulative count	4	12	18	19	21	23	24

A mean from a frequency table

$$n = \sum f, \quad \bar{x} = \frac{\sum(xf)}{\sum f}.$$

Multiply each value by how often it occurs. Averaging only the seven distinct goal counts would give each count equal weight and produce the wrong result.

(a) Percentile for 4 goals. Using the class convention of counting observations *at or below* the value,

$$\frac{4 + 8 + 6 + 1}{24}(100\%) = \frac{19}{24}(100\%) \approx 79.2\%.$$

Four goals is approximately the 79th percentile: the team scored 4 or fewer goals in about 79% of its matches. Include the match with exactly 4 goals.

(b) Mean and median.

$$\bar{x} = \frac{1(4) + 2(8) + 3(6) + 4(1) + 5(2) + 6(2) + 13(1)}{24} = \frac{77}{24} \approx 3.21 \text{ goals.}$$

For even n , average the values at positions $n/2$ and $n/2 + 1$. The cumulative counts show that the 12th value is 2 and the 13th is 3:

$$\text{median} = \frac{2 + 3}{2} = 2.5 \text{ goals.}$$

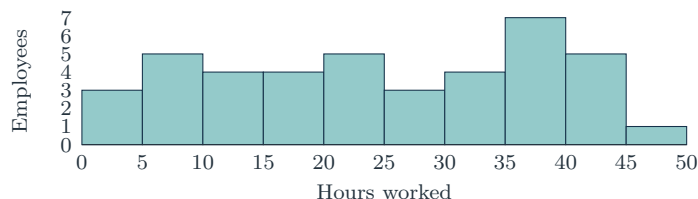
A median can be fractional even when individual observations are whole-number counts.

(c) Interpret SD. The goals scored per match typically differ from the mean of about 3.21 goals by about 2.55 goals.

(d) Remove 13 goals. The SD decreases, because the exceptionally high observation contributes a large squared distance from the mean. The remaining matches are more concentrated; the new sample SD is about 1.51 goals. Recalculate the mean and use $n = 23$ if checking numerically.

Histogram quartiles: what could the IQR be?

The 41 employees' weekly hours are grouped into five-hour intervals. Use the *cumulative frequencies* to locate ordered positions.



Hours interval	Frequency	Cumulative count	Ordered positions
[0, 5)	3	3	1–3
[5, 10)	5	8	4–8
[10, 15)	4	12	9–12
[15, 20)	4	16	13–16
[20, 25)	5	21	17–21
[25, 30)	3	24	22–24
[30, 35)	4	28	25–28
[35, 40)	7	35	29–35
[40, 45)	5	40	36–40
[45, 50]	1	41	41

Locate positions before estimating values

1. The median is the 21st observation. Exclude it, leaving two halves of 20.
2. Q_1 averages the 10th and 11th observations. Both are in the $[10, 15)$ interval, so $10 \leq Q_1 < 15$.
3. Q_3 averages the 31st and 32nd observations. Both are in $[35, 40)$, so $35 \leq Q_3 < 40$.

Using the usual bin convention shown above, the possible IQR lies between

$$35 - 15 = 20 \quad \text{and} \quad 40 - 10 = 30, \quad 20 < IQR < 30.$$

Of the choices 15, 19, 27, and 31, only **C) 27 hours** fits. For example, $Q_1 = 11$ and $Q_3 = 38$ would give $IQR = 27$ and are consistent with the bins.

Possible versus exact: The histogram does not prove that the IQR equals 27. Exact observations inside the intervals are hidden. Bin midpoints would give estimates, not exact quartiles.

Matching z-scores: strawberries and raspberries

Raspberries have mean 4.5 grams and SD 0.5 grams; strawberries have mean 15 grams and SD 1.67 grams. Find a raspberry weight with the same standardized score as a 17.5-gram strawberry.

Transfer a standardized position between two groups

$$z = \frac{x_{\text{source}} - \mu_{\text{source}}}{\sigma_{\text{source}}}, \quad x_{\text{target}} = \mu_{\text{target}} + z\sigma_{\text{target}}.$$

Here μ denotes a group's mean and σ its SD. Equivalently,

$$\frac{x_{\text{target}} - \mu_{\text{target}}}{\sigma_{\text{target}}} = \frac{x_{\text{source}} - \mu_{\text{source}}}{\sigma_{\text{source}}}.$$

Answer

The strawberry's score is

$$z = \frac{17.5 - 15}{1.67} \approx 1.497.$$

It is about 1.5 SD above its own mean. Transfer that position to raspberries:

$$x = 4.5 + \left(\frac{17.5 - 15}{1.67} \right) (0.5) \approx 5.25 \text{ grams}.$$

Thus the answer is **A) 5.25 grams**. Keep the unrounded z-score in the calculation until the last step.

Common mistake: The strawberry is 2.5 *grams* above its mean, but the raspberry must be the same number of *standard deviations* above its own mean. Use the raspberry's SD when converting back to grams.

Babysitting income: why the maximum's z-score stays fixed

Akshita earns \$12 per hour, so money earned is $Y = 12X$, where X is hours worked. Which statistic keeps the same numerical value?

Choice	Transformation	Unchanged?
A: Mean	$\bar{Y} = 12\bar{X}$	Scales by 12
B: Standard deviation	$s_Y = 12s_X$	Scales by 12
C: First quartile	$Q_{1,Y} = 12Q_{1,X}$	Scales by 12
D: Z-score of the maximum	$z_{Y,\max} = z_{X,\max}$	Yes

Answer

D) The standardized score of the maximum. Multiplying by a positive number preserves which week is the maximum, and the factors cancel:

$$z_{Y,\max} = \frac{12X_{\max} - 12\bar{X}}{12s_X} = \frac{X_{\max} - \bar{X}}{s_X} = z_{X,\max}.$$

The maximum stays the same number of standard deviations above the mean. This works for *every* corresponding observation under $Y = a + bX$ with $b > 0$ and nonzero SD.